

The empirical recurrence for OEIS A252845, recovered from its tail

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Abstract

OEIS A252845 records “Empirical recurrence of order 96” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 273$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 100$. Its last coefficient is $c_{96} = -10066329600 \neq 0$, so no further tail satisfies anything shorter; any order-96 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A252845 is “Number of $(1+2)X(n+2)$ 0..3 arrays with every consecutive three elements in every row and diagonal having one or two distinct values, and in every column and antidiagonal having two or three distinct values, and new values 0 upwards introduced in row major order.”. It has offset 1 and begins

1305, 15395, 175204, 2045858, 23788014, 277324636, 3231587589, 37668251387, ...

The entry states:

Empirical recurrence of order 96 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:41:40, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 273$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 273$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 96: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 96.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 546$ exact terms from $n = 4$ on, it returns order 96 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{96} c_i a(n-i),$$

c_1	19	c_{25}	12700228752717	c_{49}	−305375915746294614	c_{73}	1778717506931358080
c_2	−76	c_{26}	15571603739994	c_{50}	332034028958103991	c_{74}	723660330578342784
c_3	−110	c_{27}	−52297137234794	c_{51}	790132487542700634	c_{75}	−740186475373233664
c_4	−839	c_{28}	−102000796463967	c_{52}	−72540995805549962	c_{76}	−655529982834328576
c_5	7552	c_{29}	121002624338771	c_{53}	−1594737542332967717	c_{77}	288007315566286848
c_6	34038	c_{30}	464084970760516	c_{54}	−793605264796010770	c_{78}	233864230796798976
c_7	−73029	c_{31}	−133021004611231	c_{55}	2745123596701300670	c_{79}	−80842204758896640
c_8	−624023	c_{32}	−1245364899165357	c_{56}	2124400465665582328	c_{80}	−7955236231031808
c_9	−1069037	c_{33}	−276957598287514	c_{57}	−3611506964501413346	c_{81}	−37627568809775104
c_{10}	6155397	c_{34}	1468343056121339	c_{58}	−3898894117652438790	c_{82}	16040105171841024
c_{11}	28963724	c_{35}	2200060970611154	c_{59}	3368928022084608420	c_{83}	21495203796795392
c_{12}	−21412761	c_{36}	1652205275237881	c_{60}	6081984041520761812	c_{84}	−14241660081692672
c_{13}	−234930927	c_{37}	−6717742890482008	c_{61}	−1898414534664173784	c_{85}	−1421819037548544
c_{14}	−266302396	c_{38}	−9787997188116325	c_{62}	−7893228046181228912	c_{86}	4355323283111936
c_{15}	1069278997	c_{39}	11491178391958322	c_{63}	−170686436942923608	c_{87}	−2263109452759040
c_{16}	2931817350	c_{40}	13749139211563224	c_{64}	8272054736696313072	c_{88}	123404656312320
c_{17}	−4767479716	c_{41}	−1796793433577910	c_{65}	2293069472645346488	c_{89}	438856995307520
c_{18}	−2888505389	c_{42}	4037127283158157	c_{66}	−6938447855625271664	c_{90}	−137975886774272
c_{19}	1143857742	c_{43}	−38294010794068814	c_{67}	−4026815408216934832	c_{91}	−5809978212352
c_{20}	−48522399297	c_{44}	−47814547280803414	c_{68}	4569951721207834496	c_{92}	3716279173120
c_{21}	212218867351	c_{45}	69014000640588103	c_{69}	4495597403013250208	c_{93}	−244997685248
c_{22}	206769780466	c_{46}	118638548102829140	c_{70}	−2168799315476533728	c_{94}	362253647872
c_{23}	−1970482450304	c_{47}	20214437325928457	c_{71}	−3372753904085984192	c_{95}	−27313307648
c_{24}	−1673563302265	c_{48}	−237570029713519103	c_{72}	238141976264927360	c_{96}	−10066329600

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 100$, and it is the recurrence OEIS A252845 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{96} - c_1x^{95} - \dots - c_{96}$. Its constant term is $-c_{96} = 10066329600$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 96 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 96, so $\deg q = 96$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 96, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 96, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -10066329600 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-96 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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